

Checkout SVC-17 Capacity Review

13 July 2026 | Owner: Nora Vale, Checkout SRE | Next review: 2026-07-20

Decision at a glance

Decision

Hold the six-pod floor and approve two warm standby pods before Friday. This is a queue-protection decision: the release can proceed only while cache hit rate and gateway queue age remain inside their independent guardrails.

Planning peak

6,180 requests/s

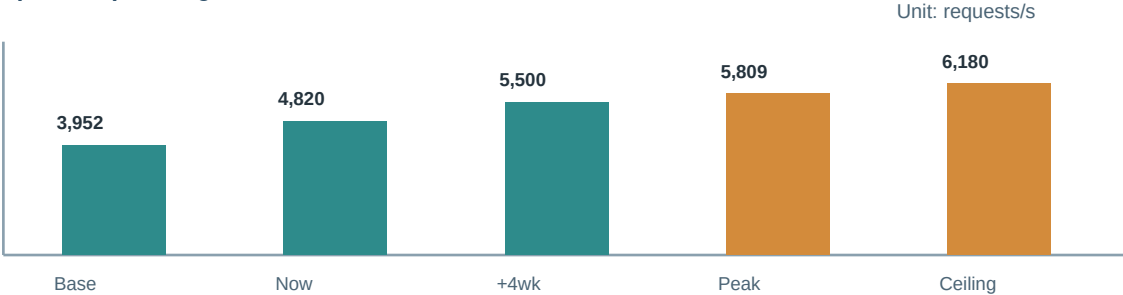
Evidence

High; incident-adjusted 15-minute series

Scope and assumptions

The review isolates the authorization path after INC-260713. It assumes the fare-release cohort reaches 1.16 times the Tuesday noon baseline, card-token cache hit rate remains above 96%, and one pod can be unavailable. The estimate excludes cart browsing and settlement retries.

Forecast profile - planning utilization



Capacity signals

Measure	Current	Planning limit	Decision signal
Authorization rate	4,820 requests/s	6,180 requests/s	Add warm pods when three intervals exceed 5,560.
Token cache hit	96.8%	>= 95.5%	Cache erosion, not pod count, is the first escalation.
Gateway queue age	18 ms	< 65 ms	Freeze promotions if queue age crosses the guardrail.
Checkout error share	0.7%	< 1.2%	Protect the remaining error budget 2.4%.

Plan, controls, and ownership

Action plan

Action	Due	Owner	Acceptance evidence
Warm pool	Jul 15	Checkout SRE	Pin two standby pods and verify 90-second admission.
Fare rehearsal	Jul 17	Release manager	Exercise the release mix through synthetic payment stubs.
Cache drill	Jul 19	Platform cache	Remove one cache shard and record recovery slope.
Go/no-go	Jul 20	Nora Vale	Approve only if queue age stays below 65 ms.

Risk register

Risk	Likelihood	Impact	Response
Issuer latency burst	Medium	High	Route to held tokens before scaling the web tier.
Cache shard loss	Low	High	Keep a hot replica with a 12-minute promotion target.
Release cohort overlap	Medium	Medium	Stagger fare-release groups by ten minutes.

Decision rationale

Hold the six-pod floor and approve two warm standby pods before Friday. This is a queue-protection decision: the release can proceed only while cache hit rate and gateway queue age remain inside their independent guardrails.

Selective cross-references

Review alongside: Checkout demand forecast and message bus throughput plan. These linked plans are relevant because they share a limiting dependency or coordinated operating window.

Model notes: all values are synthetic planning figures. No customer records, production identifiers, secrets, or routable network addresses are included. Northstar Transit Cloud | Checkout SVC-17 | 2026-07-13 | error budget 2.4% | INC-260713